ARYAN CHANDRA

Pittsburgh, PA | (412) 606-5002 | aryanc2@andrew.cmu.edu | linkedin.com/in/aryan-chandra-aa24411ba | github.com/aryan-chandra-cmu | portfolio

Robot-learning researcher building motion-imitation, retargeting, and reinforcement-learning pipelines for custom legged robots. Experience spans kinematics, parallel PPO training, curriculum learning and domain randomization, sim-to-real validation, and real-time embedded policy deployment.

TECHNICAL SKILLS

Reinforcement & Robot Learning: PPO, imitation learning, reward shaping, domain randomization, curriculum learning, sim-to-real transfer, locomotion policy training and evaluation
Simulation & ML: MuJoCo, MJLab, rsl_rl, PyTorch, NumPy
Robotics: legged locomotion, kinematics, dynamics, inverse kinematics, contact dynamics, embedded motor control
Programming & Systems: Python, C++ (Arduino), Linux, Git, real-time robot control, Arduino UNO Q, Dynamixel, IMU integration

RESEARCH EXPERIENCE

Research Assistant | Robomechanics Lab, Carnegie Mellon University Fall 2025 – Present
NSF EAGER-funded research on learning-based locomotion for bio-inspired quadrupeds

- Developed automated training and validation tooling for motion extraction, retargeting, IK refinement, contact analysis, constraint verification, and rollout evaluation; diagnosed failures in coordinate frames, actuator mappings, linkage constraints, and contact behavior.
- Trained PPO-based velocity-tracking locomotion policies in MuJoCo/rsl_rl across 2,048 parallel environments; designed rewards plus velocity-command, domain-randomization, and rough-terrain difficulty curricula for robust sim-to-real locomotion.
- Deployed PPO locomotion policies to physical Microtaur hardware using a dual-rate Arduino UNO Q controller, with Linux-based inference and a real-time microcontroller loop for IMU/joint feedback and 8-motor Dynamixel control through asynchronous state/action buffering.
- Designed controlled morphology experiments across rigid and spine-enabled quadrupeds, holding policy architecture, reward structure, actuator configuration, and action space constant; demonstrated an 11% reduction in mean landing force with a passive roll spine while preserving forward locomotion.

SELECTED PROJECTS

Generalized Imitation Learning and Retargeting for Legged Robots | CMU Summer 2026

- Built an end-to-end MuJoCo/MJLab pipeline to transfer 3D canine locomotion across four custom robot morphologies using closed-chain kinematics, morphology-aware inverse kinematics, gait synchronization, and contact-aware grounding.

Waypoint Tracking Using Central Pattern Generators and PPO | CMU

- Developed a PPO-based waypoint-tracking controller for a SpotMicro quadruped in MuJoCo, integrating a central pattern generator for stable trot gaits and goal-directed steering toward CSV-defined targets; contributed environment design, reward shaping, and training-pipeline implementation.

WORK EXPERIENCE

GET and Automation Intern | Schneider Electric Jan 2024 – Aug 2025

- Analyzed warehouse retrieval workflows and designed automation improvements for MLSS and goods-to-person systems, reducing floor-space requirements by 35%, required workforce by 50%, and improving bin-placement accuracy by 39.5%.

EDUCATION

Carnegie Mellon University | M.S. Mechanical Engineering, Robotics and Control Dec 2026
Manipal Institute of Technology | B.Tech Mechanical Engineering Sep 2024